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SPECIFIC
TECHNICAL

Gas measurement systems

Hourly gas measuring devices

Part 7: Remote management systems for gas meters

UNI/TS 11291-7

May 2013

Gas measurement systems

Hourly based gas metering systems

Part 7: Gas meters remote management system

The technical specification specifies the construction and functional requirements of the Concentrator, Repeater, SAC and Translator devices, defined in UNI/TS 11291-1 and used in the remote management system aimed at the remote reading and remote management of gas measuring devices serving the Delivery Points of the distribution networks.

ITALIAN TEXT

This technical specification replaces UNI/TS 11291-7:2011.

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PREMISE

Compared to the previous edition, the regulatory references have been updated, alignment with UNI EN 12405-1 has been carried out, point 5.2 for the mechanical and electrical characteristics and point 9 on the use cases of the concentrator have been added.

The technical specification is reviewed every three years. Any comments on UNI/TS 11291-7 must be received by UNI by May 2015.

This technical specification was developed under the jurisdiction of the body federated to UNI

CIG - Italian Gas Committee

The UNI Central Technical Commission gave its approval on 23 April 2013.

This technical specification was ratified by the President of UNI and became part of the national regulatory body on 23 May 2013.

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1 PURPOSE AND SCOPE

This technical specification specifies the construction and functional requirements of the Concentrator, Repeater, SAC and Translator devices, defined in UNI/TS 11291-1 and used in the remote management system aimed at the remote reading and remote management of gas measuring devices serving the Delivery Points of the distribution networks.

2 REGULATORY REFERENCES

This technical specification refers, through dated and undated references, to provisions contained in other publications. These normative references are cited at the appropriate points in the text and are listed below. For dated references, subsequent amendments or revisions made to those publications are valid only if introduced into this technical specification as an update or revision. For undated references, the latest edition of the referenced publication (including updates) applies.

UNI/TS 11291 (all parts)	Gas Metering Systems - Hourly Gas Metering Devices
IEC EN 60529	Degrees of protection of enclosures (IP Code)
IEC EN 62056-21	Electricity metering - Data exchange for meter reading, tariff and load control - Part 21: Direct local data exchange

3 TERMS AND DEFINITIONS

For the purposes of this technical specification, the terms and definitions set out in UNI/TS 11291-1 apply.

4 SAC

4.1 General description

The SAC function in the remote reading or remote management system is the part of the center responsible for managing the relevant network.

All centers that have the right to communicate with the GdMs in any capacity must do so, unless otherwise provided, exclusively through the SAC, which is the only component of the system delegated to this function.

The SAC function can be physically separate from the rest of a center's functions (data storage, analysis, editing, validation, etc.) or it can be integrated.

A remote reading or remote management network can have multiple SACs.

The SAC function in the remote reading or remote management system is responsible for managing the network in terms of:

- configuration/reconfiguration of all network devices (Concentrators, GdMs, Repeaters, Translators);
- automatic and periodic acquisition of data processed by the GdMs via direct connection (PP4: CTR, CTE, DLMS) or by the Concentrators (PP3) using all the variants provided by the distributor (push, inbound, outbound, TCP IP);
- routing procedures towards the GdMs through Concentrators, Repeaters and Translators;
- acquisition of events and alarms processed by all devices on the network;
- synchronization of the clocks of the directly managed GdMs and of the Concentrators.

The SAC must implement mechanisms for the authentication and identification of directly connected Concentrators, Translators, and GdMs.

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The data exchanged between SAC and Concentrators, Translators and GdM, unless otherwise specified, must always be encrypted and authenticated.

The SAC must be able to perform all the services defined by UNI/TS 11291-1 that are relevant to it.

The SAC must also oversee the following procedures:

- formal validation of first-level data (valid, invalid, or unreliable), interpreting the events of individual devices, excluding the commercial validation of the data value (the SAC does not enter into the merits of the data);
- activation of Concentrators, Translators, and GdMs during installation. The SAC awaits the operator's completion of the field configuration and declares the remote management activation process complete;
- provision of a statistical diagnostic of GdM coverage (reachable unreachable).

4.2 Purpose

The SAC is not among the architectural elements of the remote management/remote reading system for which full interoperability is required. Full interoperability between the elements of the remote management system must therefore be enabled between the SAC and the GdM; the methods by which the functionalities described in section 4.3 are implemented are therefore not specified. To ensure system-wide interoperability, however, it is necessary to provide implementation instructions for some of the SAC's functionalities and services to avoid differing interpretations. It is also necessary to specify which information managed by the SAC must be made available to the incoming distributor in the event of a concession change. The following sections provide instructions for ensuring system-wide interoperability.

4.3 Minimum functionality

4.3.1 General information

In order to perform the services indicated in UNI/TS 11291-1, the SAC must execute certain procedures consisting of a series of organized and structured functions (over time). These procedures are initiated by internal automation or are the result of high-level commands originating from outside the SAC (for example, an operator or another system).

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prospectus 1 **Functionality vs. Devices**

Command	Internal External	A2	A1	TO	B	C	R	T2	T4	THERE IS
Synchronize date and time	*THE*	x	x	x	x	x	-	-	x	x
Configure metrological parameters	*AND*	x	x	x	x	x	-	-	-	-
Configure business parameters (for example: pricing schedule, billing period)	*AND*	x	x	x	x	x	-	-	-	-
Take charge of a new apparatus (enlistment)	*I/E*	x	x	x	x	x	x	x	x	x
Configure routing	*THE*	-	-	-	-	-	x	x	x	x
Decommission a device	*AND*	x	x	x	x	x	x	x	x	x
Suspend an apparatus	*I/E*	x	x	x	x	x	x	x	x	x
Restore default values	*I/E*	x	x	x	x	x	x	x	x	x
Reset event queue	*THE*	-	-	-	-	-	-	x	x	x
Activate/end the maintenance state of a device	*THE*	-	-	-	-	-	x	x	x	x
Change a profile's privileges	*AND*	-	-	-	x	x	-	-	-	-
Change the security parameters of a device	*I/E*	x	x	x	x	x	x	x	x	x
Change SAC communication references	*THE*	(x)	x	x	x	x	-	x	x	x
Change the service SLA	*AND*	x	x	x	x	x	-	-	-	x
Suspend/Re-enable supply	*AND*	x	-	-	-	-	-	-	-	-
Send message to user	*AND*	x	x	-	-	-	-	-	-	-
Firmware update	*AND*	x	x	x	x	x	x	x	x	x
Capture events	*THE*	x	x	x	x	x	x	x	x	x
Acquire Configuration	*THE*	x	x	x	x	x	x	x	x	x
Get instant data	*THE*	x	x	x	x	x	-	-	-	-
Get tariff data	*I/E*	x	x	x	x	x	-	-	-	-
Acquire anti-fraud logs	*THE*	x	x	x	x	x	x	x	x	x
Acquire quality of service logs	*THE*	x	x	x	x	x	x	x	x	x
Acquire profiles (flow, pressure, etc.)	*I/E*	x	x	x	x	x	-	-	-	-
Acquire auxiliary logs	*I/E*	x	x	x	x	x	-	-	-	-
Network Discovery	*I/E*	-	-	-	-	-	x	x	x	x
Acquire routing	*I/E*	(x)	(x)	(x)	-	(x)	(x)	x	x	x

Legend
THE = process activated internally to the SAC. = process
AND activated by an entity external to the SAC.
A2,A1,A,B,C= Classes of GdM as specified in UNI/TS 11291-1. R
= Repeaters.
T2 = Translators for PM2 protocol (see UNI/TS 11291-1). =
T4 Translators for PP4 protocol (see UNI/TS 11291-1). =
THERE IS Concentrator.
x = function that affects the type of device and is performed directly.
- = not applicable.
(x) = function performed indirectly through other devices in the network.

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4.3.2 Clock synchronization

The SAC must synchronize the clocks of the components managed in direct mode (GdM, Concentrators and Translators) of the networks under its jurisdiction.

Synchronization must be able to occur at every connection to the device.

The synchronization of the device clocks must be carried out in such a way as to guarantee the synchronization of the device clocks, with respect to universal time, within the values set by UNI/TS 11291-1 and by the other parts of the technical specification applicable to each device.

To ensure a high level of precision in the synchronization of the clocks of the devices, the SAC must synchronize its own clock while maintaining a maximum error, with respect to the UTC date and time, of no more than 1 s.

Synchronization of the devices must occur every 7 days at the latest.

In order to highlight problem areas related to the synchronization of the equipment clocks, the SAC must report with a Warning in the relevant On-Board Log, the GdMs and the equipment with which it has not communicated for more than 7 days.

4.3.3 Dynamic multi-tariff capacity

The GdMs must be able to process gas consumption across structured tariff bands (tariff programs), and it must be possible to configure and reconfigure the tariff programs, even remotely.

A tariff program (PT) is a collection of related information defined by the Authority and useful to the GdM for managing gas consumption tariffs. It is characterized by:

- an identifier (ID_PT) that uniquely identifies the PT, [for example, constructed on the basis of the resolution issued by the AEEG]. For interoperability reasons, it is represented by two bytes, the first referring to the year and the second representing a progressive number from 1 to 255 (for example: 09:25 is the 25th PT in the year 2009); the value 0 is reserved to indicate the absence of the tariff program. The same ID, regardless of the SAC, identifies the same PT, so the ID value must be shared and normalized;
- an explanatory notes field that adds customizable clarifications to the PT;
- the Billing Period field which identifies the frequency with which the GdM must manage the closing of the billing period (divisors of 12);
- the effective date field, which defines the date, expressed in year, month, and day, on which the PT entered or will enter into force;
- the PT descriptor field, which defines the time periods, seasonal and weekly, during which the rate bands are in effect. The following rules apply:
    - the days of the week must be divided into weekdays, Saturdays and holidays (holidays are Sundays, the national holidays indicated in table 2 and up to 15 configurable days);

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prospectus 2

January 1st
January 6th
Easter Monday According to the Gregorian Calendar
April 25th
May 1st
June 2nd
August 15th
November 1st
December 8th
December 25th
December 26th

- There are two weekly programs, one for the summer season and the other for the winter. Each weekly program has an effective month/day that repeats throughout the season and each year;

Note The first program ceases to be in effect when the second begins.

- for each type of day, a maximum of 5 alternations are possible, delimited by (whole) hours within which one of three possible tariff bands is in force (indicated as F1, F2, F3).

- the default band descriptor field, which defines the band in which consumption must be recorded in the event of a lack of time synchronization.

Periodically (at least once a month) the ID value of the PT in force in the GdM must be checked or the absence of the tariff program.

In operation, a PT must be able to be configured in the GdM installed in a PdR[1)] Configuration can occur directly (point-to-point communication) or indirectly through Translators or Concentrators or locally using a terminal.

4.3.4 Measurement and consumption profiles

The SAC must activate functions to acquire the measurement data and consumption profiles processed by the GdMs.

The type of data that must be acquired by the SAC depends on the type of GdM and is indicated in Table 3.

The frequency with which this data must be acquired depends on the configured SLA (Service Level Agreement).

4.4 Service SLA

Service Level Agreements (SLAs) establish what minimum data must be acquired by the SAC and at what intervals.

In SLA 0, the minimum service level, the SAC is able to trace the volumes at reference thermodynamic conditions in the relevant time bands from the relevant consumption profiles and determine consumption in the previous billing period based on the absolute totals of the volumes at reference thermodynamic conditions recorded at the end of each conventional gas day. It can also manage the switches using the same data. The data required for SLA 0 must therefore be acquired every day. The totals of the volumes at the alarm measurement conditions can only be acquired when an alarm is present.

1) PdR is the gas delivery point in the distribution networks.

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In SLA 1, the GdM is managed using the information it records regarding the absolute consumption totals differentiated by time band and billing period. The acquisition frequency may be limited to the end of the billing period. Consumption profiles may be acquired as needed.

SLA 2 is an extension of SLA 1 with the addition of consumption profiles. In SLA 2, the SAC must acquire data daily.

SLA 3 and SLA 4 are complex service levels that require daily data capture by the SAC.

Table 3 summarises the SLAs and the related applicable data in relation to the GdM class.

prospectus 3

Classes of devices	A2	A1	TO	B	C	ALS				
Data						0	1	2	3	4
Absolute totalizer of volumes at the reference thermodynamic conditions of the current billing period	*M*	*M*	*M*	*M*	*M*		x	x		x
Absolute total of volumes at the reference thermodynamic conditions recorded at the end of the day		*M*	*M*	*M*	*M*	x			x	x
Absolute totalizer of volumes at the reference thermodynamic conditions of the previous billing period	*M*	*M*	*M*	*M*	*M*		x	x		x
Absolute totalizer of volumes at the reference thermodynamic conditions of the current billing period and in the tariff bands (F1, F2, F3)*)	*M*	*M*	*M*	*M*	*M*		x	x		x
Absolute volume totalizers at alarm measurement conditions		*M*	*M*	*M*	*M*	x	x	x	x	x
Absolute totalizers of volumes at alarm measurement conditions in the tariff bands (F1, F2, F3)*)				*M*						x
Absolute totalizer of volumes at the reference thermodynamic conditions of the previous billing period and in the tariff bands (F1, F2, F3)*)	*M*	*M*	*M*	*M*	*M*		x	x		x
Conventional flow rate at max reference conditions of the day			*OR*	*OR*	*OR*				x	x
Cumulative diagnostics over the period	*M*	*M*	*M*	*M*	*M*	x	x	x	x	x
Hourly flow profiles at reference thermodynamic conditions	-	*M*	*M*	*M*	*M*	x		x	x	x
Hourly flow profiles at measurement conditions	-	-	*OR*	*M*	*M*				x	x
Daily volume profiles at reference thermodynamic conditions	*M*	-	-	-	-	x		x	x	x
1-hour mean pressure profiles	-	-	-	*M*	-					x
1-h average temperature profiles				*OR*	-					x
Flow profiles at measurement conditions, flow at reference conditions, pressure, temperature averaged over 15 min	-	-		-	*OR*					x

Legend
*) Only if a tariff program is active. =
M mandatory.
OR = optional in relation to the SLA.
- = not required.

4.5 Information needed in case of takeover

4.5.1 General information

In order to enable a new concessionaire to take over the remote management network, the transferring distributor must make available a series of information regarding the structure and status of the network at the time it is about to change operator.

All information must be transmitted securely.

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4.5.2 Network structure

In order to describe the information that the transferor must make available to the incoming transferor that will allow him to transfer the remote reading network, the formalism defined at international level must be used[2)], as shown in Table 4.

prospectus 4

Given	Attributes	Notes
Identifier	PdR (for GdM) Serial number (for other devices)	It also allows you to trace the physical location managed by the asset management
Network address	MAC address IP Telephone number Other	Referring to Service Delivery Points
Communication windows		Referring to Service Delivery Points
Safety	Algorithms Password Keys	Referring to Service Delivery Points
Configuring the Concentrators	Update date List of affiliated SDPs	Referring to Concentrator-type Service Delivery Points
Configuring Translators	Update date List of affiliated SDPs	In the case of GdM

4.5.3 Network status at the time of delivery (Onboard Log)

The SAC manages a register called the On-Board Register (RdB) in which events occurring in the network or networks it manages must be recorded.

All events that occur during the operation of the SAC and that concern interactions with the field devices must be recorded in the on-board log.

The Logbook is one of the pieces of information that must be delivered to the incoming Distributor in the event of a concession switch.

The information recorded in the RdB must be distinguished, in relation to its significance, into:

- Simple EVENT: event recorded for notification purposes (for example: New GdM registered);
- WARNING: an event that must be brought to the attention of an operator that alters but does not affect the functioning of the system or equipment (for example: incomplete firmware download);
- ALARM: event that must be brought to the attention of an operator as it affects the functioning of the system or of a device (example 1: ID_PT in use in the GdM other than the official one; example 2: anomaly in the shut-off valve of a GdM-A2).

The records in the RdB must contain at least the information indicated in Table 5.

2) This formalism is defined by DTMF (Distributed Management Task Force), in document DSP 0004.

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Field	Information
1	Progressive
2	Event Code
3	Date and time (with details to the second) of recording
4	Date and time (with details to the second) of occurrence (if possible to detect)
5	Date and time (with details to the second) of notification (if applicable)
6	Type (Event/Warning/Alarm)
7	Source device (GdM, Repeater, Translator, Concentrator, SAC)
8	Source reference (PdR, Serial number) if applicable
9	Process reference (synchronization, configuration, diagnostic, etc.) if applicable
10	Status (to be notified/notified)
11	Identifier of the operator who provided the notification
12	Event Description

The SAC must be able to maintain in the RdB at least the events relating to the last 6 months of management.

5 CONCENTRATORS

5.1

General information

The data concentrator is the peripheral device designed to collect metering group sampling data and transmit it to the remote management center (SAC). In the application context, it is the device that performs at least the following functions:

- the management of a local database;
- the management of at least one level of communications towards the SAC;
- the management of at least one level of Point-to-Multipoint communications for communication towards the GdMs, using for this purpose any Repeaters to extend radio coverage and any Translators to translate the protocols present on the meters;
- the management of at least one local point-to-point communication level for local configuration and programming activities via a maintenance terminal.

In particular, the Concentrator must:

- allow the SAC to have access to all the minimum functions offered by gas meters;
- manage the exchange of information on the underlying wireless network;
- make available to the SAC, whenever requested, the updated periodic data, collected by the GdMs in the appropriate time windows;
- provide the SAC with complete visibility of the operating configuration of the controlled wireless network, informing it of any updates that have occurred;
- ensure system interoperability between peripheral devices and SACs from different manufacturers.
- ensure the correct execution of identification, authentication, and access authorization mechanisms for both the GdM and the SAC and maintenance terminals. These mechanisms are those defined with reference to UNI/TS 11291-10 and the respective protocols defined in the relevant sections of the UNI/TS 11291 series.
- carry out verification activities and, if necessary, restore the integrity of stored data, highlighting any unreliability of the data.

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5.2 Mechanical and electrical characteristics

The mechanical protection rating must be at least IP55 in accordance with IEC 60529. Its weight must be such as to facilitate installation, which must be possible for a single operator, even in different positions. The mounting must be such as to allow it to be installed either on a wall or against a pole.

The Concentrator must operate correctly in the ambient temperature range defined by the manufacturer, chosen in the range between -10 °C and +55 °C.

The Concentrator can be powered by:

- *low voltage electrical network 230 Vac 50 Hz*In this case it must provide a secondary power source (buffer battery). The battery must guarantee:
  - a useful life of not less than 5 years in the operating temperature range and in all operating conditions,
  - an autonomy of 7 days, in case of lack of energy supplied by the electrical network; during this period all functions must be guaranteed,
  - a charging time of less than 6 hours starting from a completely flat battery;
- *solar panel and buffer battery*The solar panel and battery must be sized to ensure a useful life of the panel and battery of no less than 20 years and 5 years, respectively, within the operating temperature range and under all operating conditions; a battery life of 7 days in the event of insufficient or prolonged absence of panel lighting; and a battery charging time of less than 6 hours from a completely flat battery.
- *batteries*: of any technology provided that they guarantee the Concentrator an autonomy of at least 5 years under the following reference operating conditions:
  - in the operating temperature range,
  - active communication, towards SAC, for at least 2 minutes per day,
  - communication with up to 128 devices (GdM, Repeaters).

In relation to their management capacity, 2 profiles of concentrator devices can be identified:

- L profile: capable of managing at least 128 devices between GdM and repeaters;
- H profile: capable of managing at least 1024 devices between GdM and repeaters.

5.3 Concentrator Features

5.3.1 General information

The main functions of the Concentrator can be classified as:

- features related to communications management;
- features related to application procedures;
- other features.

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Below is a list of functions and who originates them.

Functions (Procedures)	Function originated from
Periodic data reading	Concentrator and/or GdM
Instant reading of data	SAC
Gas Disconnection	SAC
Gas Reconnection	SAC
Sending Messages to the End Customer	SAC
Commissioning	SAC
Configuration	Concentrator and/or SAC
Network Analysis	Concentrator and/or SAC
Synchronization	Concentrator
Firmware update	SAC
Self-diagnostics	Concentrator

The functions listed above can be performed by sending commands to the devices affiliated with the Concentrator; the commands can be:

- collective commands (broadcast), i.e. intended for a multitude of devices (if supported by the protocol);
- individual commands, i.e. intended for a single device.

The Concentrator, with regards to the management of affiliated devices, must support both commands that require confirmation from each individual recipient (confirmed), and commands that do not require confirmation from each individual recipient (unconfirmed).

Collective commands, if supported by the protocol, can affect all devices affiliated with the Concentrator or only different types with possibly other selective criteria (for example, when downloading the firmware, the manufacturer, the type, the software version).

5.3.2 Data collection and storage

The Concentrator carries out periodic or occasional data collection activities for the devices affiliated with it.

The purpose of periodic data collection is to update the database of affiliated devices in the Concentrator, periodically check the status of the devices and make this information available to the SAC asynchronously and independently of the actual availability of communication with the affiliated devices.

To this end, the Concentrator must periodically collect, at a configurable interval, at least by device class, gas consumption information from each affiliated GdM and maintain, on a non-volatile memory device, a data image of each GdM, for both the current and previous tariff periods. The Concentrator must also store the device's summary status, which includes at least the following information: the associated PdR identifier, diagnostic status, number of active events, tariff program identifier, and installed software identifier.

It must also store and manage the encryption/authentication keys used by each affiliated device on the portion of the network affected by PM1, as prescribed in UNI/TS 11291-10.

When configuring affiliated devices, which can be initiated by the SAC asynchronously with respect to the actual availability of communication windows to the devices, the Concentrator must be able to store the set of parameters to be sent to each type of affiliated device in a rewritable, non-volatile memory area, retaining this data at least for the time necessary to successfully complete the operation. For this purpose, the SAC must associate a validity date with each command intended for a device, after which the Concentrator deems the command no longer forwardable.

The time interval at which the Concentrator has access to the data of the affiliated devices is defined by the validity period of the information (for example, tariff periods) or by the Service Level or by the supervision and maintenance needs of the network and the affiliated devices.

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Table 6 shows the associated collection time interval for each data type.

5.3.3 Periodic data reading

The communication that allows for the periodic reading of data processed by the underlying devices (GdM, Translators, and Repeaters) can originate from these same devices (push mode) or from the Concentrator (pull mode) and must in any case be carried out within the time windows in which communication with the affiliated devices is active. The Concentrator must therefore be able to support both modes.

The data acquired through the periodic reading process must be stored and maintained securely (see UNI/TS 11291-10) in a dedicated non-volatile memory area dedicated to all types of affiliated devices (both GdM and Translators and Repeaters). The date and time of the last acquisition must also be recorded.

For each stored data item, the Concentrator must provide evidence of its validity to the SAC or authorized terminal. To this end, each recorded data item must be accompanied by a qualifier.

In the case of GdMs, the frequency of reading, as well as the nature and quantity of the data, may be different for the different classes of GdMs (see table 6).

For other affiliated devices, such as Translators and Repeaters, the Concentrator must periodically acquire at least the diagnostic log (see points 6 and 7).

prospectus 6

GdM Class			Register
A2	A1	TO	
x	x	x	Absolute total of volumes converted at the reference conditions in the current billing period
x	x	x	Absolute total of volumes converted under the reference conditions in the current billing period and relating to bands F1, F2 and F3
x	x	x	Absolute total of volumes converted under the reference conditions in the previous billing period
x	x	x	Absolute total of volumes converted under the reference conditions in the previous billing period and relating to bands F1, F2 and F3
- -	1)	1)	Absolute total of measured volumes in error in the current period
x	2)	2)	Absolute totalizer of volumes converted to the reference thermodynamic conditions in error
- -	1)	1)	Absolute total of volumes measured in error in the previous billing period
x	2)	2)	Absolute totalizer of volumes converted to the reference thermodynamic conditions in error in the previous billing period
x	x	x	Current diagnostics
x	x	x	Number of events recorded
x	x	x	Anti-fraud registers
x	x	x	Cumulative diagnostics for the previous billing period
x	x	X	Date, time (and reason) of the closure of the current tariff period
x	x	x	Rate program ID in use in the current billing period
x	x	x	Rate program ID used in the previous billing period
x	x	x	Identifier of the software in use and any software waiting to be activated
x	- -	- -	Gas flow shut-off valve status

1) If measured volumes are managed.
2) Alternatively, the measured volumes will be in error if the latter are not managed.

Each of the values of the registers listed in Table 6 must be associated with a qualifier and a time reference, referring to the moment in which the value was detected on the GdM.

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5.3.4 Instant reading of data

The extemporaneous collection of GdM data is performed by the Concentrator at the request of the SAC or an authorized terminal. Collection may occur for at least the following reasons:

- acquisition of measurement data to manage a contractual variation;

- real-time verification of the status of the generic affiliated device (GdM, Repeater, Translator);

- check the configuration of the generic affiliate device;

- acquisition of the flow profiles of the GdMs (hourly or daily depending on the GdM class);

- acquisition of diagnostic or alarm events from generic affiliated devices.

Since the communication network to the GdMs may not be available when the SAC receives one or more requests for impromptu reading, the Concentrator must store such requests in a job queue and then process them at a deferred time when the communication network is available. It must also make available a rewritable non-volatile memory area sufficient to store the data associated with the requests (for example, acquisition of hourly or daily profiles).

To this end, the Concentrator must be able to store data resulting from a request queue at least 10% deep than the number of devices that can be affiliated with the Concentrator, regardless of the type of data request. These pending requests are stored in a queue managed according to the "FIFO" (First In, First Out) logic, with the same associated priority. Each request must be numbered and must indicate its associated priority, the date it was made by the SAC, the date it was processed, and the date it was completed (by response or timeout) by the Concentrator. The data associated with each request must include the date and time the data update was produced and the data qualifier.

The Concentrator must be able to manage at least 3 priority levels associated with the different types of extemporaneous requests (see table 6).

When pending requests can be processed, the responses received from the generic affiliated device must be numbered to ensure association with the relevant request. Data must be stored in a non-volatile memory area designated for this purpose, while the associated events must be stored in the relevant queue. Both must be kept available to the SAC until requested. Once the data associated with an impromptu request has been successfully transferred to the SAC, the associated memory area can be made available to process subsequent requests. The sequence of events associated with the requests and their status are always queued until the memory buffer assigned to that queue is full. In this case, and as soon as possible, the Queue Full event must be notified to the SAC, after which subsequent events are overwritten.

Each time the SAC requests a new series of queries, the Concentrator checks the consistency between the list of recipients present in the request and the list of affiliated devices, thus rejecting activities that could keep the request itself in an indefinite pending state.

The SAC must be informed, whenever it requests it, of the processing status of the requests themselves: completed, pending, GdM does not respond.

To carry out its functions, the Concentrator must have sufficient memory to store all the data associated with the periodic collection of all the affiliated devices it can manage, for at least a period of 40 days.

For GdMs, the following data indicated in table 7 can be acquired extemporaneously.

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prospectus 7

GdM Class			Register
A2	A1	TO	
	x	x	Hourly flow profile converted to reference conditions
x			Daily flow profile converted to reference conditions
x	x	x	Quality of service logs (battery time remaining)
x	x	x	Anti-fraud logs (clock synchronization totalizers, number of unauthorized accesses)
x	x	x	Device personal information (firmware identifiers, serial number, etc.)
x	x	x	Network parameters
x	x	x	Flow profile at conventional maximum daily reference conditions (with time of occurrence)
x	x	x	Events
x			Supply Management
x			Messaging

For Repeater and Translator devices, the following data indicated in table 8 can be acquired extemporaneously.

prospectus 8

Apparatus		Register
R	T2, T4	
x	x	Service quality logs (battery remaining time, clock synchronization totalizers)
x	x	Anti-fraud logs (number of unauthorized accesses)
x	x	Device personal information (firmware identifiers, etc.)
x	x	Network parameters
x	x	Events

5.3.5 Configuration

In the case of configuration commands intended for the Concentrator itself, the command can come from the SAC (remote configuration) or locally via a terminal connected to the relevant logic port.

During the remote/local communication phase relating to the configuration of the Concentrator, all the functions of the Concentrator itself must be kept active.

During the configuration phase, the Concentrator must switch from the Operational state to the Maintenance state and this state transition must be recorded in the Concentrator's event log, reporting the date and time of the configuration.

The parameters that can be configured must be kept in a non-volatile memory medium and checked periodically for any alterations (see point 5.3.9).

The Concentrator must allow the configuration of at least the following parameters[3]:

- network address and/or telephone numbers;
- encryption/authentication keys for communications on ports P1 and P3;
- access password for communications on ports P1 and P3;
- other safety parameters;
- communication windows to port P1 and P2;
- GdM descriptors (personal data, PdR, encryption keys, passwords, routing, ...);
- translator descriptors (personal data, encryption keys, passwords, routing, ...);
- Repeater descriptors (personal data, encryption keys, passwords, routing, ...).

3) For the definition of ports P1, P2 and P3, please refer to UNI/TS 11291-1.

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5.3.6 Network Discovery

The main objective of the network analysis is to update the network configuration by identifying any other devices that may be or need to be affiliated with the Concentrator.

The function can be activated by non-affiliated devices and/or periodically by the Concentrator or requested by the SAC. For this purpose, the PP3 protocol[4]It provides objects for configuring and/or activating the function and the Concentrator must be able to support both functions.

In the network analysis process, the Concentrator also involves any Repeater and Translator devices already affiliated.

5.3.7 Synchronization

The clock synchronization of the affiliated devices must be performed starting from the Concentrator clock at each connection, using the mechanisms provided by the PM1 protocol and implementing the appropriate algorithms for correcting the delays introduced by the repetitions.

The Concentrator must be equipped with a calendar clock device capable of calculating time with adequate precision (drift < 1 min per month).

It must also automatically manage Daylight Saving Time, which must be able to be deactivated.

The Concentrator's real-time clock (RTC) must be periodically synchronized by the SAC, so that the residual error with respect to the local time is contained within a maximum permissible deviation (MDA) of 2 s.

Self-synchronization mechanisms (e.g. from GPS or NTP) are permitted as long as the deviation following synchronization remains within the limits indicated above (MDA).

The Concentrator's clock synchronization must be performed autonomously by the Concentrator itself, in immediate or diluted mode (recovering 2 s every minute) with the following rules:

- if the synchronization involves a shift in the Concentrator time within the limits of the MDA, the synchronization is not carried out;
- if the synchronization involves a time shift greater than MDA and up to 2 hours, the synchronization is performed by generating an event;
- if the synchronization involves a time shift of more than 2 hours, the synchronization must not be possible and an alarm must be generated.

The clock must not undergo variations or malfunctions in the event of failure or incorrect reception of the synchronization command.

For battery-powered concentrators, the internal clock must function for at least 2 minutes in the event of battery replacement; for other concentrators, the same function must be maintained in the absence of power for at least 7 days.

5.3.8 Firmware update

The firmware (FW) update process of the underlying devices sees the Concentrator as a pure executor of the procedure activated by the SAC.

To this end, the Concentrator must be able to reserve at least a memory buffer adequate for the maximum size of the FW for updating the software of the underlying devices.

The SAC must therefore indicate to the Concentrator which devices the software update is intended for by providing the following information:

1) Manufacturer (code).

2) Type of device (GdM, Repeater, Translator).

3) Device or Model or Class (A2, A1, A, B, C, T2, T4; unique identifier of the device).

4) For definitions, please refer to UNI/TS 11291-1.

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4) Firmware version to update.

5) Downloading index in one-to-one correspondence with the version and monotonically increasing.

6) Date by which the GdM must implement the switch to the new firmware version.

7) Name of the executable file.

Since the software update procedure of the underlying devices by the Concentrator is a procedure that could require much more time than that permitted by the time window dedicated to PM1, the FW update of the devices can take place by successively forwarding code segments of an appropriate length dictated by the PM1 protocol.

For details on implementing the procedure, please refer to UNI/TS 11291-10.

5.3.9 Self-diagnostics

The Concentrator must be equipped with a function capable of carrying out the checks reported in Table 9.

prospectus 9

Verify	Frequency	Notes
Database integrity	Every time it is used	The result of the check must be reported in the diagnostic log and in case of corrupted integrity an event must be generated.
How communication channels work	Every time it is used and in any case at least once a day	Failure must be recorded in the events
The status of affiliated devices	For every communication	
The state of the power source	At least every 1 min	Absence and restoration must be recorded in the events
The battery charge status (if applicable)	Sampled to detect variations in residual charge of up to 1% of the initial charge	- A diagnostic warning should be activated when the remaining charge is less than 20% - An automatic shutdown circuit must be provided when the battery charge is exhausted

5.4 Concentrator Database

The Concentrator must have a database located on a non-volatile memory medium in which the parameters indicated in Table 10 must be recorded for each managed device.

prospectus 10

Parameter	GdM	Translator	Repeater	Notes
Typology	X	X	X	
Builder	X	X	X	
Model/Class	X	X		A2, A1; A, B, C, T2, T4
Firmware version and ID	X	X	X	
Date and time of affiliation	X	X	X	
Associated Routing	X	X	X	
Date and time of last communication	X	X	X	
Device diagnostics	X	X	X	
Communication quality logs	X	X	X	
Affiliate hierarchy	X	X	X	Direct, Translator address, Repeater address
State	X	X	X	Affiliate; Hasn't communicated for at least "t" time; Needs affiliation.

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The Concentrator must make the following data available to the authorized SAC or terminal upon request:

- Concentrator Type (L, H).
- Manufacturer.
- Firmware version in use.
- Firmware version to be activated (with activation date).
- Network address.
- Telephone numbers (GSM, SMS, PSTN, etc.).
- Number of affiliated GdMs.
- Number of affiliated Repeaters.
- Number of affiliated Translators.
- Diagnostics.
- Remaining battery charge time (for battery-powered concentrators only).
- Anti-fraud logs (login attempts with incorrect encryption key, login attempts with incorrect password).
- Number of events recorded.
- Events; at least the following: prolonged primary power failure, synchronization, software update, prolonged absence of communication to the SAC.

6 REPEATERS

6.1 General characteristics

Repeaters are auxiliary network elements capable of extending radio communication coverage between GdMs and Concentrators; they process information necessary for communication but do not store data processed by the GdMs.

The Repeater can communicate, using PM1 class protocols, with the GdMs, with the Concentrator and with other Repeaters.

Their task is to retransmit messages correctly received with the PM1 protocol without examining the context of the data contained in the message and intended for the application layer; it therefore operates only at the "data link" and "network" protocol layers.

6.2 Mechanical and electrical characteristics

The Repeater must operate correctly in the ambient temperature range defined by the manufacturer, chosen in the range between -10 °C and +55 °C.

To ensure the interoperability of the Repeaters, the technical characteristics of the Repeater, when they do not coincide with the GdM, must comply with the following:

- at least IP55 protection level; its weight must be such as to make it easy to install, which must be possible for a single operator, even in different positions.

- fixing: both wall and pole fixing must be provided.
- diet:
    - *from low voltage mains 230 Vac 50 Hz*In the case of the presence of a secondary power supply (buffer battery), the battery must guarantee:

      a) a useful life of not less than 5 years in the operating temperature range and under all operating conditions,

      b) an autonomy of 7 days, in case of absence of energy supplied by the electrical network,

      c) a charging time of less than 6 hours starting from a completely flat battery;

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- *from solar panel and buffer battery*; the solar panel and buffer battery must be sized to ensure a battery life of no less than 5 years within the operating temperature range and under all operating conditions; an autonomy of at least 7 days, in the event of insufficient or prolonged absence of panel illumination; and a battery charging time of less than 6 hours starting from a completely flat battery.

  The maximum dimensions allowed for the photovoltaic panel are: 300 mm× 400 mm (b×h).

- *from batteries*, of any technology provided that they guarantee an autonomy of at least 5 years under the following reference operating conditions:

  - in the operating temperature range;

  - lower communication active for at least 3 minutes per day;

  - repetition of 5 apparatuses.

6.3 Additional requirements

The reception sensitivity is consistent with that expected by PM1.

The power of the retransmitted signal must be modulable and compliant with the PM1 standard.

The maximum number of Repeaters (”hops”) to reach the most remote GdM starting from the Concentrator must be limited in accordance with the PM1 profile protocol and in any case such as not to compromise the minimum battery life.

In case a GdM acts as a Repeater, its batteries must be sized in such a way as to maintain the same standard operating duration expected for GdMs.

The number of GDMs underlying a single Repeater must be such as not to compromise battery life when evaluated in the most frequent communication scenarios permitted in UNI/TS 11291-1.

To manage the communication windows specified in UNI/TS 11291-1, the Repeater must be equipped with a clock function capable of classifying at least the type of day (Monday-Sunday) and guaranteeing a maximum drift of 2 minutes/month. The Repeater's clock must be synchronized by the Concentrator or other authorized device. The residual clock error after synchronization must be maintained within 2 seconds. Self-synchronization is permitted as long as it guarantees the above-mentioned requirements.

The synchronization command received (for example from the Concentrator) must be replicated to the network elements underlying the Repeater itself (GdM or other Repeaters), using its internal clock in order to avoid errors due to propagation times in multiple synchronizations.

The retransmission time of the command or received data must comply with the requirements defined by PM1 technology to avoid excessive latencies and therefore uncontrolled shifts in the activation time windows.

The Repeater must be enrolled by the SAC via the Concentrator or the Translator. Message routing must be implemented only on authenticated network elements.

The Repeater applies any encryption to the commands addressed to it (for example, defining time slots) and not to repeated messages (in which the encryption remains end-to-end).

The Repeater could have an active role in creating a route as envisioned by PM1, but it could also be an executor of a route determined by a centralized algorithm.

The duration of the Repeater's wake-up time windows and their opening time must be configurable. To simplify installation and maintenance operations, the Repeater can manage one or more dedicated and configurable wake-up windows for this purpose.

The Repeater must allow for local and remote firmware updates.

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The Repeater must process and make available a diagnostic that collects at least the following information:

- the time synchronization error (> 4h);
- battery warning below 20% of the total charge;
- other alarms (generic fault, corrupted database).

The Repeater must process and make available upon request the following information:

- device serial number;
- network address;
- date and time of the internal clock;
- remaining battery time expressed as a % of the total duration declared by the manufacturer;
- the strength level of the received signal;
- any other information suitable for reconstructing the network topology;
- version of the active firmware and of any firmware that needs to be activated;
- diagnostic logs (anti-fraud, synchronization, etc.).

The above parameters must be retained in memory even in the absence of power.

For battery-powered Repeaters, the internal clock must remain operational for at least 2 minutes if the battery is replaced; for other Repeaters, the same function must be maintained in the absence of power for at least 48 hours.

The manufacturer must provide a method for activating communication on an extemporaneous basis for installation and/or maintenance purposes of the Repeater.

7 TRANSLATORS (GATEWAY)

7.1 General characteristics

Translators are auxiliary network elements capable of enabling protocol translation, but do not store data processed by the GdMs.

The Translator performs a protocol translation between PM1 class protocols and PP4 or PM2 class protocols.

In relation to the ability to translate protocols, Translators can be of two types:

- T2 - Translators capable of translating the PM1 protocol into PM2 and vice versa;
- T4 - Translators capable of translating the PM1 protocol into PP4 and vice versa.

T2 Translators must be able to perform protocol translation at all levels of the protocol stack, including the application layer, and for this reason, they must be able to encrypt and decrypt messages received and to be translated. To this end, T2 Translators must know the encryption keys of all their associated GdMs. The time delay introduced must be compatible with the timeframes required by the encryption itself.

T4 translators, on the other hand, perform protocol translation only at the lower layers of the stack (physical, data link, and network), which are not encrypted. The data received by the sending device and contained in the application layer are packaged in a new protocol stack, known to the receiving device, without the translator decoding or interpreting its contents. The sending device is responsible for encoding and encrypting the application layer data using rules known to the receiving device.

The Translator of any type applies encryption to the commands intended for it (for example, defining time slots).

The Translator can play an active role in creating a route as required by PM1.

For energy saving reasons, the Translator, of any type, communicates only during time windows that can also be configured remotely as established by UNI/TS 11291-1.

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The number of GdMs underlying a single Translator must be such as not to compromise the battery life evaluated in the most frequent communication cases permitted in UNI/TS 11291-1.

7.2 Mechanical characteristics

The mechanical protection rating must be at least IP54 according to IEC 60529. Its weight must be sufficient to facilitate installation, which must be possible for a single operator, even in different positions. The mounting must be such that it can be installed either on a wall or against a pole.

7.3 Electrical characteristics

The translator must operate correctly in the ambient temperature range defined by the manufacturer, chosen in the range between -10 °C and +55 °C.

The Translator can be powered by:

- *low voltage electrical network 230 Vac 50 Hz*In this case, a secondary power source (buffer battery) must be provided; the battery must guarantee:
  - a useful life of not less than 5 years in the operating temperature range and under all operating conditions,
  - an autonomy of 7 days, in case of lack of energy supplied by the electrical network,
  - a charging time of less than 6 hours starting from a completely flat battery;

- *solar panel and buffer battery*The solar panel and buffer battery must be sized to ensure a battery life of no less than 5 years within the operating temperature range and under all operating conditions; a battery life of 7 days in the event of insufficient or prolonged absence of panel illumination; and a battery charging time of less than 6 hours from a completely flat battery.

  The maximum dimensions allowed for the photovoltaic panel are: 300 mm×400 mm (b×h).

- *batteries*of any technology as long as they guarantee the Translator an autonomy of at least 5 years in the following reference operating conditions:
  - in the operating temperature range;
  - communication to the SAC (if applicable) active for at least 2 min every 3 days;
  - communication with a maximum number of 10 devices (GdM, Repeaters and Translators) for a total of 2 min/d.

7.4 Other functions

To manage communication windows, the Translator must be equipped with a clock function capable of classifying at least the type of day (Monday-Sunday) and guaranteeing a maximum drift of 120 s/month, without synchronization. The Translator's clock must be synchronized remotely (for example, by the SAC or the Concentrator) or locally; the residual clock error after synchronization must be kept within 2 s.

In order to avoid errors due to propagation times in multiple synchronizations, the synchronization command, for example received from the SAC, must be replicated by the Translator to the GdMs, using its internal clock.

To avoid excessive latencies and therefore uncontrolled shifts in the activation time windows, the retransmission time of the command or data received must not exceed the values indicated by PM1 technology.

The Translator must be authenticated directly by the SAC or Concentrator; message routing must be performed only on authenticated network elements.

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The Translator must process and make available a diagnostic containing at least 1 status word that collects at least the following information:

- time synchronization error (> 4 h);
- battery warning below 20% of the total charge;
- other alarms (generic fault, corrupted database).

The Translator, upon request of the SAC or the Concentrator or the authorised Terminal, communicates:

- your personal data [serial number, manufacturer, model, class of the device (T2,T4)];
- network address;
- the supported communication protocols;
- date and time of the internal clock;
- the remaining battery life expressed in percent;
- the number of unauthorized login attempts;
- the firmware version (software identifier) you are using and the firmware version you are about to activate;
- the PdRs and network addresses of the GdMs/Repeaters/Translators affiliated with it and for each of them the routing (series of device addresses) used to reach it;
- the reception signal level of the GdM/Repeater/Translator/Concentrator in direct communication;
- the level of the reception signal from the SAC (for T4 type translators);
- Diagnostic logs (anti-fraud, number of recorded events, events (at least the following: prolonged power failure, synchronization, software update, prolonged absence of communication to the SAC), etc.).

The essential parameters of the Translator such as its network address, any routing tables, and the time slot descriptor, must be kept in memory even during battery replacement.

For battery-powered translators, the clock function must be maintained for at least 2 minutes during battery replacement; for other translators, the same function must be maintained in the absence of power for at least 48 hours.

The Translator must allow for local and remote firmware updates.

The manufacturer must provide a method for activating communication on an extemporaneous basis for installation and/or maintenance purposes of the Translator.

The Translator does not go into the specifics of requests coming from the SAC and directed to the affiliated devices (whether they are periodic or impromptu) but must be able to manage them correctly in terms of the quantity of data and priority levels associated with the requests themselves and is not obliged to decipher the application content of the response messages received.

Logical inconsistencies in SAC requests, as well as incorrectly encrypted messages, are verified by the receiving device and rejected with a message that states, in an appropriately encoded manner, the reason for the rejection so that the outcome of a command or query is always available to the SAC.

8 TERMINALS

8.1 Purpose

The remote management system devices manufactured in compliance with this technical specification require, in some phases of their life cycle and operation, to undergo local interventions to be carried out through interaction with a terminal available to the operator.

These terminals also require a sufficient degree of interoperability to ensure that a single terminal is able to operate on devices of the same type and GdM of the same class, even if from different manufacturers.

The following points provide the requirements necessary to achieve this level of interoperability.

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8.2 General characteristics

It is considered necessary to have a terminal that interacts locally with the field devices in the following operating phases:

- first configuration: when the device is installed and put into operation; in this phase the terminal must be able to provide the device with all the information necessary for communication with the rest of the relevant network and the parameters necessary for operation;
- maintenance: during repairs, restorations, battery replacements, metrological and/or functional checks;
- island operation: intended only for GdMs, to perform ordinary process functions (readings, configurations, user management, etc.) in the absence of a connection to the network or to the SAC.

The terminal must be equipped with the interfaces necessary for local connection with the type of device for which it is designed.

The terminal must be able to operate on battery power with a minimum operating autonomy of 2 h and maintain data persistence on mass storage devices even in the absence of primary power for at least one month.

The weight of the terminal must not exceed 7 kg including all accessories.

The terminal must have its own unique identifier in the distribution network.

The terminal must be equipped with a calendar clock which must be synchronizable to ensure a maximum error of 10 s with respect to universal time.

Terminals must support at least one of the following physical interfaces that can be addressed to different system devices.

prospectus 11

Apparatus	GdM					Translators	Repeaters	Concentrators
Class	A2	A1	TO	B	C			
Interface								
RS232	-	-	x	x	x	x	x	x
Optics to)	x	x	x	x	x	x	x	x
PP4	x	x	x	x	x	x		
PM1	x	x	x	-	-	x	x	x

to) Complies at least for the physical layer with IEC 62056-21.

Terminals must be able to manage at least 2 usage profiles, one of which is intended for an administrator.

8.3 Functional requirements

The Terminal must guarantee at least the functions indicated in the following tables (from 12 to 15), in relation to the type of device to which it is connected and in compliance with the safety requirements referred to in UNI/TS 11291-10.

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prospectus 12

GdM	Installation	Maintenance	Island Mode
Date/Time Synchronization	x	x	-
Diagnostic Cancellation	x	x	-
Clear event log	-	x	
Change status (normal, maintenance, suspended, etc.)	x	x	-
Support for GdM checks	x	x	-
Support for verification of solenoid valve actuation (if applicable)	x	x	-
Enabling the solenoid valve to reopen	-	x	-
Communication verification support	x	x	-
Battery replacement support	-	x	-
Programming of functional parameters	x	x	-
Programming communication parameters	x	-	-
Programming tariff parameters (optional)	x	x	x
Reading tariff data	-	-	x
Reading configuration	-	-	x
Reading statistical registers (communication, anti-fraud, etc.)	-	-	x
Configuring user messages	-	-	x
Software update	-	x	-

prospectus 13

Concentrator	Installation	Maintenance
Date/Time Synchronization	x	x
Diagnostic Cancellation	x	x
Clear event log	-	x
Change status (normal, maintenance, suspended, etc.)	x	x
Initializing the internal database	x	-
Support for communication verification, with the SAC	x	x
Battery replacement support (if applicable)	-	x
Programming of functional parameters	x	x
Programming communication parameters	x	x
Reading configuration parameters	x	x
Read the list of affiliated GdMs, the list of affiliated Translators, and the list of affiliated Repeaters	-	x
Reading statistical registers (communication, anti-fraud, etc.)	x	x
Software update	-	x

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prospectus 14

Translator	Installation	Maintenance
Date/Time Synchronization	x	x
Diagnostic Cancellation	x	x
Clear event log	-	x
Change status (normal, maintenance, suspended, etc.)	x	x
Initializing the internal database	x	-
Support for communication verification, with the SAC/Concentrator	x	x
Battery replacement support (if applicable)	-	x
Programming of functional parameters	x	x
Programming communication parameters	x	x
Reading configuration parameters	x	x
Reading the list of GdM affiliates	-	x
Reading statistical registers (communication, anti-fraud, etc.)	x	x
Software update	-	x

prospectus 15

Repeater	Installation	Maintenance
Diagnostic Cancellation	x	x
Clear event log	-	x
Change status (normal, maintenance, suspended, etc.)	x	x
Initializing the internal database	x	-
Support for communication verification, with the Concentrator	x	x
Battery replacement support (if applicable)	-	x
Programming of functional parameters	x	x
Programming communication parameters	x	x
Reading configuration parameters	x	x
Reading the list of GdM affiliates	-	x
Software update	-	x

9 CONCENTRATOR USE CASES

prospectus 16

Data

Req. ID	Text
[Req. CD001]	The Concentrator must be able to configure the GdM with wake-up windows. The allocated time window is configurable for each meter.
[Req. CD002]	The configuration of the GdM wake-up windows must be done using an algorithm that ensures a wake-up distribution that is not critical to the network load. The algorithm must be modifiable by the SAC (e.g., t0 + frequency, Random, etc.).
[Req. CD003]	In the event that the Concentrator functions with wake-up windows, the Concentrator's wake-up windows must be consistent with the wake-up windows of the GdMs it manages.
[Req. CD004]	Upon enlistment, the Concentrator must send all configuration commands that the SAC has foreseen, then providing the SAC with feedback on the outcome of those commands.
[Req. CD005]	In case of failure in sending commands to the GdM, the Concentrator must provide retry logic, before giving feedback to the SAC on the task failure.
[Req. CD006]	The frequency of the transfer to the SAC must be configurable based on the PP3 network loads.

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prospectus 17 **Event collection**

Req. ID	Text
[Req. EC001]	The Concentrator must store all responses sent by the GdMs regarding the outcome of the configuration commands received (tasks).
[Req. EC002]	Confirmation of receipt of commands must be made available to the SAC, with the references of the command to which it refers.
[Req. EC003]	The Concentrator must be able to request the data contained in the Event Register and the device diagnostic data (status word) from the GdMs, forwarding them to the SAC.
[Req. EC004]	The Concentrator must be able to generate events internally, which will then be reported to the SAC, when particular situations occur, including: - synchronization; - prolonged lack of primary power supply; - software update; - prolonged absence of communication with the SAC; - internal DC alarms (power supply, generic fault, etc.); - maintenance interventions on the DC; - tampering attempts (when detectable).

prospectus 18 **GdM configuration (Device Configuration)**

Req. ID	Text
[Req. DC001]	The Concentrator must send all configuration commands it receives from the SAC to the GdM, providing the latter with feedback on their outcome. The commands that can be sent to the GdM and received by the SAC are at least: - commercial configurations (Price Bands/Billing Period); - technical configurations (wake-up windows); - “on demand” reading commands; - switch commands; solenoid valve opening/closing commands.
[Req. DC002]	The Concentrator is responsible for synchronizing the GdM clock. In turn, the Concentrator's clock is synchronized by the SAC.
[Req. DC003]	The Concentrator must be able to perform software updates on the GdMs, in the manner provided by each manufacturer.
[Req. DC004]	The Concentrator must support the "SCAN" function, which identifies GdMs transmitting within its radio coverage range. The SCAN function is activated by the SAC when needed.

prospectus 19 **Network Configuration**

Req. ID	Text
[Req. NC001]	The Concentrator must store and update the configuration of the PM1 network underlying it (for example by providing the list of “affiliated” GdMs) with information on the radio channel and the quality of the received signal.
[Req. NC002]	The Concentrator must make information and services available to the SAC so that the SAC can proceed with the analysis, optimization and configuration of the networks underlying the concentrators.
[Req. NC003]	The Concentrator must be able to manage only devices authorized by the SAC.
[Req. NC004]	The concentrator receives from the SAC all the information necessary for the enlistment of the GdMs to be managed.
[Req. NC005]	The concentrator must keep the list of visible GdMs updated, complete with information useful for network management by the SAC.
[Req. NC005]	The concentrator must maintain the list of GdMs enrolled by the SAC.
[Req. NC006]	In case of changes to the routing settings, the concentrator must receive from the SAC the list (or updates to this list) of GdMs not to be managed.

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20 Monitoring

Req. ID	Text
[Req. RT001]	The Concentrator must provide the SAC with useful information so that the SAC can analyze the quality of communication between the devices and the Concentrator, in order to identify any anomalous operating conditions that need to be managed appropriately.
[Req. RT002]	The Concentrator must make its alarms available to the SAC.
[Req. RT003]	In the event of failure to receive GdM data, the Concentrator must recover them using the methods configured by the SAC.

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